

Literature Review: gPPM - A Generalized Matrix Operation and Parallel Algorithm for Erasure Code Encoding/Decoding

1. Background and Motivation

Erasure codes are used in large storage systems (cloud storage, data centers) to protect against disk, sector, and node failures while using less storage overhead than simple replication. The paper divides erasure codes into two families:

- Symmetric Parity codes (e.g., Reed-Solomon/RS, Cauchy RS, STAR) - every parity block is computed from the same number (k) of data blocks.
- Asymmetric Parity codes (e.g., SD code, PMDS code, LRC code) - different parity blocks are computed from different numbers of data blocks (e.g., local parity vs global parity in LRC, or disk parity vs sector parity in SD code). These newer codes were designed to handle real failure patterns better (like combined disk + sector failures, or fast degraded reads), but existing implementations still decode them using the traditional parity-check matrix method, which is slow, serial, and treats all lost data as one big block to recover together.

The authors' key observation: some lost blocks are actually independent of each other, they can be recovered separately, in parallel, without waiting on other lost blocks. Existing decoding methods miss this and process everything serially.

2. Contribution Overview

The paper proposes two algorithms:

- PPM (Partitioned and Parallel Matrix) - works only on asymmetric parity codes. It splits the parity-check matrix H into smaller sub-matrices based on which lost blocks are independent, decodes the independent parts in parallel across threads, and picks a cheaper calculation sequence (order of matrix multiplication) to reduce the number of operations.
- gPPM (generalized PPM) - extends PPM's idea to also work on symmetric codes like RS, by dynamically choosing, per-matrix, whichever calculation order (normal vs. matrix-first) needs fewer actual multiply/XOR operations. This makes it more broadly applicable and squeezes out further gains even inside PPM's sub-matrices.

This is a journal extension of an earlier ICPP 2015 conference paper (PPM only); gPPM is the new material.

3. Methodology

Step 1 - Cost model

The authors count operations in terms of `mult_XORs()` (multiply a block by a Galois-Field constant, then XOR into an accumulator). Given the decoding equation $BF = F^{-1} \cdot S \cdot BS$, there are two ways to compute it:

- Normal sequence: multiply $S \cdot BS$ first, then apply F^{-1} .
- Matrix_first sequence: multiply $F^{-1} \cdot S$ first (a small matrix $\times$ matrix product), then apply to BS .

Which is cheaper depends on the number of nonzero elements produced by each approach - this varies by code and parameters, so no single sequence always wins.

Step 2 - Exploit independence (PPM)

Using a log table derived from the parity-check matrix, the algorithm finds which lost blocks depend only on themselves (or a small self-contained group) versus which are entangled with everything else. Independent groups become separate small sub-matrices; the remaining sub-matrix handles the entangled dependent blocks.

Step 3 - Parallelize

The independent sub-matrices are decoded concurrently across up to T threads (T capped by core count to avoid overhead), then the recovered values feed into decoding the remaining sub-matrix.

Step 4 - Fine-grained sequence selection (gPPM)

Rather than statically picking one calculation order for the whole matrix (as PPM does), gPPM computes the cost of both sequences per sub-matrix (accounting separately for the relative cost of multiply vs. add operations) and picks the cheaper one on the fly. This also lets it apply the same trick to RS/GRS codes, which have no independent block structure to exploit via partitioning, only sequence choice.

Evaluation setup

Implemented in C on top of an open-source SD encoder/decoder, using Intel SIMD (ISA-L-style) acceleration, tested across three CPUs (4/6/8 cores), varying stripe size, number of disks (n), data/parity counts (k , m), sector counts (r , s , z), and thread count (T).

4. Key Results

- PPM improves SD/LRC code encode/decode speed by up to 210.81% (avg ~61% for SD, 16–37% for LRC).
- gPPM improves further over PPM by up to 102.41% in about a third of tested SD configurations (PPM remains optimal in the rest).
- gPPM improves plain RS code by up to 32.25%, and the RS variant used inside SD code (GRS) by up to 79.73%.
- Improvements hold across different CPUs, thread counts, and stripe sizes, confirming the gains come from reduced computation, not just hardware-specific tuning.
- Power/energy overhead from added threads is small (~2 W).

5. Related Work Positioning

The authors distinguish their approach from:

- New code designs (EVENODD, RDP, STAR, SD, LRC, EC-wide) - these change what is computed, not how it's computed.
- SIMD/GPU acceleration (Intel ISA-L, GPU-based decoding) - these speed up individual operations, and are complementary (used underneath PPM/gPPM in the experiments).
- APCM - also reduces computational cost via matrix transformation, but optimizes matrix-matrix operations (cheap) rather than matrix-vector operations against actual data blocks (expensive), so its real-world gains are limited.
- Other parallelism schemes (block-level, disk-level, node-level, equation-oriented) - PPM/gPPM is described as a matrix-oriented parallelism approach, distinct from these because it exploits the algebraic structure of independent faulty blocks in H itself.

6. Conclusion of the Paper

PPM/gPPM together give a general, code-agnostic technique to speed up erasure code encoding/decoding by combining (a) exploiting parallelism among independent lost blocks and (b) dynamically choosing the cheaper matrix computation order - applicable to both older symmetric codes (RS) and newer asymmetric codes (SD, LRC, PMDS).

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Chen, J., Yang, S., Wang, Y., Ye, M., Lei, F. (2024) - Data repair accelerating scheme for erasure-coded storage system based on FPGA and hierarchical parallel decoding structure, Cluster Computing, 27(6): 7803–7823. The

cited paper uses this paper's check-matrix-based decoding approach as part of an FPGA hardware acceleration design for Cauchy Reed-Solomon (CRS) code repair.

Sequential Baseline Implementation and Results

Milestone 1 outcome

The scalar reference implementation is complete for the full published SD parameter grid. All 7,833 representative feasible layouts recover correctly, and all 180 timed 32 MiB trials pass validation.

Implementation Scope

The implementation follows the traditional parity-check-matrix process in Section 2.2 of Li et al. It builds H , extracts faulty columns F and surviving columns S , inverts F over $GF(2^w)$, and reconstructs the missing sectors with the normal calculation sequence. Encoding uses the same solver with the parity locations treated as the unknown set.

$$T = S \times B_S \quad B_F = F^{-1} \times T$$

The baseline supports $GF(2^8)$, $GF(2^{16})$, and $GF(2^{32})$. It is intentionally single-threaded and scalar: PPM partitioning, OpenMP, SIMD, matrix-first evaluation, and dynamic gPPM selection are excluded so later milestones have an unambiguous reference.

Correctness Evidence

The Figure 2 example reproduces $u(S) = 22$, $u(F^{-1}) = 13$, and $C = 35 = C1$. The encoded stripe satisfies $H \times B = 0$, erased sectors are reconstructed byte-for-byte, and the recovered stripe again satisfies $H \times B = 0$.

The full parameter-grid sweep loads all 3,969 published FAST coefficient sets and combines each with every z in $1 \leq z \leq s$. It tests one deterministic representative fault layout per feasible (configuration, z) point: the leftmost m disks and s extra sectors spread over exactly z rows. This covers every parameter tuple, but does not enumerate every combinatorial fault placement.

Measure	Result
Published coefficient configurations	3,969
Total (configuration, z) points	7,938
Geometrically infeasible points	105
Representative feasible layouts tested	7,833
Successful encode/decode round trips	7,833 / 7,833
Singular matrices / over-counts / count errors	0 / 0 / 0
Exact C1 matches	7,649 (97.65%)
Benign results below C1	184 (2.35%)

The 184 under-counts are coefficient cancellations, not failures. In those layouts, entries of F^{-1} become zero, reducing the actual number of region operations below the paper's generic-position expression. The measured count still equals $u(S) + u(F^{-1})$ in every tested layout.

Baseline Benchmark

The compact benchmark fixes $n = 16$, $r = 16$, $z = 1$ and evaluates every m, s in $\{1, 2, 3\}$ pair. Each codeword is exactly 32 MiB, comprising 256 sectors of 128 KiB. One warm-up precedes ten encode and ten decode trials per configuration. Random generation, erasure injection, validation, checksums, and CSV output are outside the timed interval.

Item	Recorded setting
Processor	Intel Core Ultra 9 185H
Compiler	GCC 11.4.0, -O2
Execution	One thread, scalar, no SIMD
Code parameters	$n=16, r=16, z=1$; all m, s in $\{1, 2, 3\}$
Codeword / sector size	32 MiB / 128 KiB
Trials	1 warm-up + 10 measured per operation
Primary metric	Useful non-parity MiB/s

m	s	Field	C	Encode MiB/s	Decode MiB/s
1	1	$GF(2^8)$	527	852.60	854.58
1	2	$GF(2^{16})$	783	634.68	631.51
1	3	$GF(2^{32})$	1039	673.01	671.77
2	1	$GF(2^8)$	828	444.98	447.81
2	2	$GF(2^{16})$	1084	399.48	401.21

m	s	Field	C	Encode MiB/s	Decode MiB/s
2	3	GF(2 ³²)	1340	479.27	476.19
3	1	GF(2 ⁸)	1159	282.56	275.55
3	2	GF(2 ¹⁶)	1415	276.56	279.68
3	3	GF(2 ³²)	1671	352.91	349.51

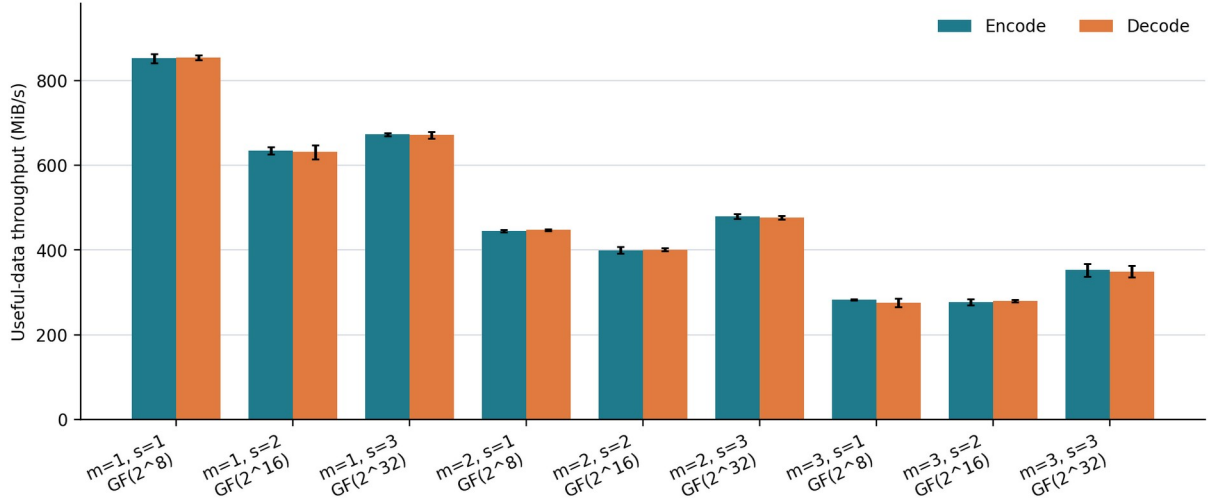


Figure 1. Mean sequential useful-data throughput; error bars show one standard deviation across ten trials.

All 180 measured trials pass byte comparison, syndrome, and operation-count validation. Throughput falls as parity work increases because more nonzero matrix coefficients are applied to each stripe. Field width alone does not determine speed because the three scalar field backends use different table organizations. These local values are not directly comparable with the paper's SIMD results on older processors; they are the baseline for this repository.

Reproducibility and Scope

make test - Run arithmetic, parser, construction, and GF(2³²) tests.

make run - Reproduce the Figure 2 smoke test.

make sweep - Regenerate full parameter-grid correctness results.

make benchmark - Regenerate the 180 timed baseline trials.

Detailed evidence is retained in sweep_results.csv and benchmark_results.csv. The next milestone should compare an OpenMP PPM implementation against exactly these correctness checks, configurations, data sizes, and throughput definitions.